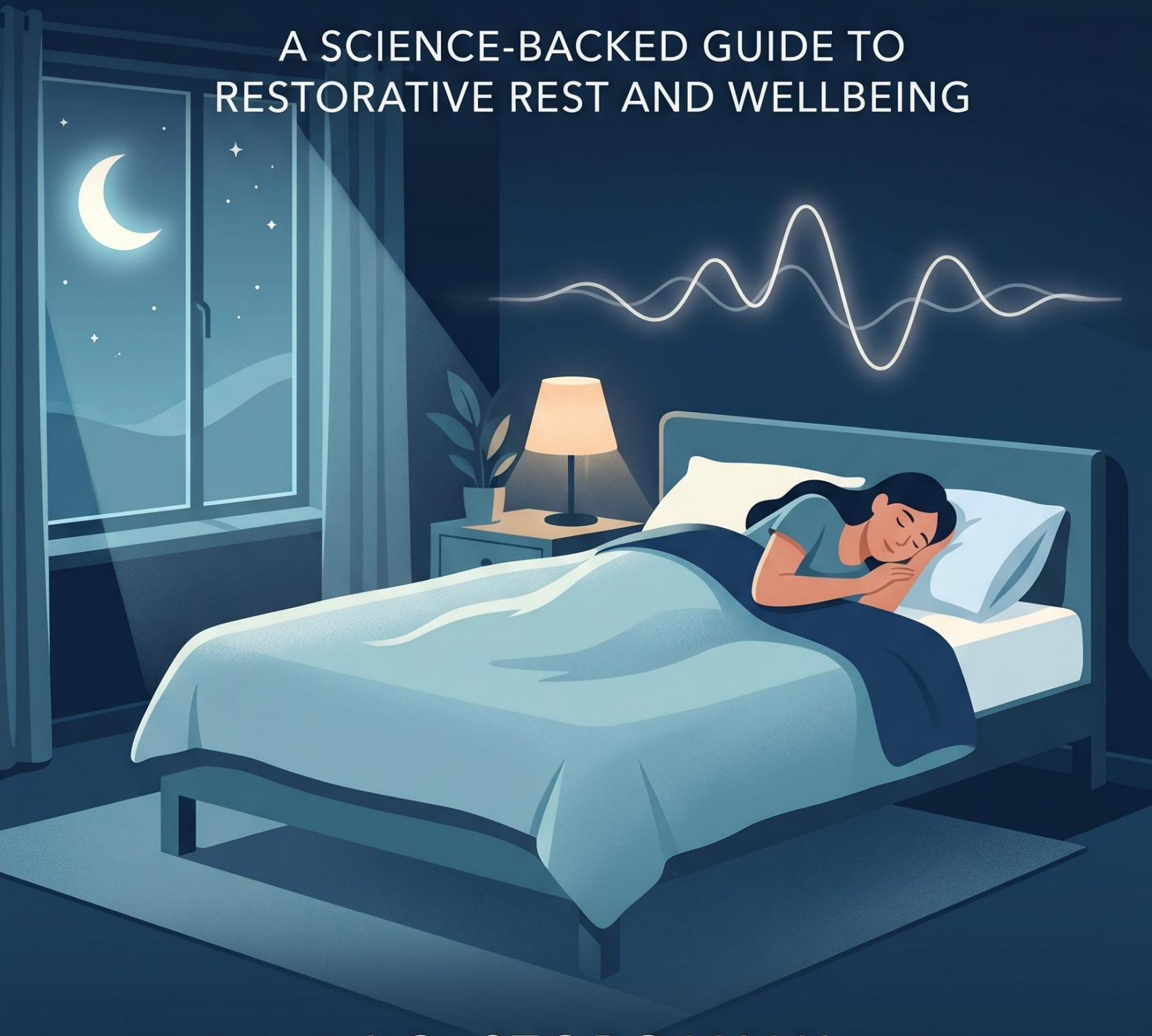


DEEP SLEEP OPTIMIZATION

A SCIENCE-BACKED GUIDE TO
RESTORATIVE REST AND WELLBEING



J.C. STORGANAU

DEEP SLEEP OPTIMIZATION

A Science-Backed Guide for Real People

Why you wake up exhausted — and the practical system to fix it

28-Day Action Plan Included

How to Use This Guide

You can read this guide cover to cover in a few hours, but the real value comes from the next 28 days. Each chapter ends with a short, specific action — one change you can make tonight or this week. Stack those changes gradually and most people feel a meaningful difference within two to three weeks.

Week	Focus	Daily Actions
Week 1	Light & Timing	Anchor your circadian clock with morning light and a fixed wake time.
Week 2	Environment & Routine	Optimise your sleep space and build a wind-down habit.
Week 3	Food & Caffeine	Adjust meal timing and set a caffeine cut-off.
Week 4	Supplements & Data	Add supplements if needed; review weekly trends, not nightly scores.

This guide does not replace medical care. If you suspect sleep apnea, chronic insomnia, or another health condition, use it alongside — not instead of — professional support.

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PART I

Understanding Deep Sleep

1 Why You're Exhausted — and Why It's Not Your Fault

You went to bed at a reasonable hour. You clocked something close to eight hours. Yet you woke up foggy, reached for coffee before you'd brushed your teeth, and felt vaguely behind for the rest of the day. Sound familiar?

You are far from alone. A large proportion of adults in industrialised countries report regularly waking unrefreshed despite spending adequate time in bed. The core problem is usually not how long you sleep, but how deeply. And the forces eroding that depth — artificial light, irregular schedules, late caffeine, chronic stress — are woven into the texture of modern life.

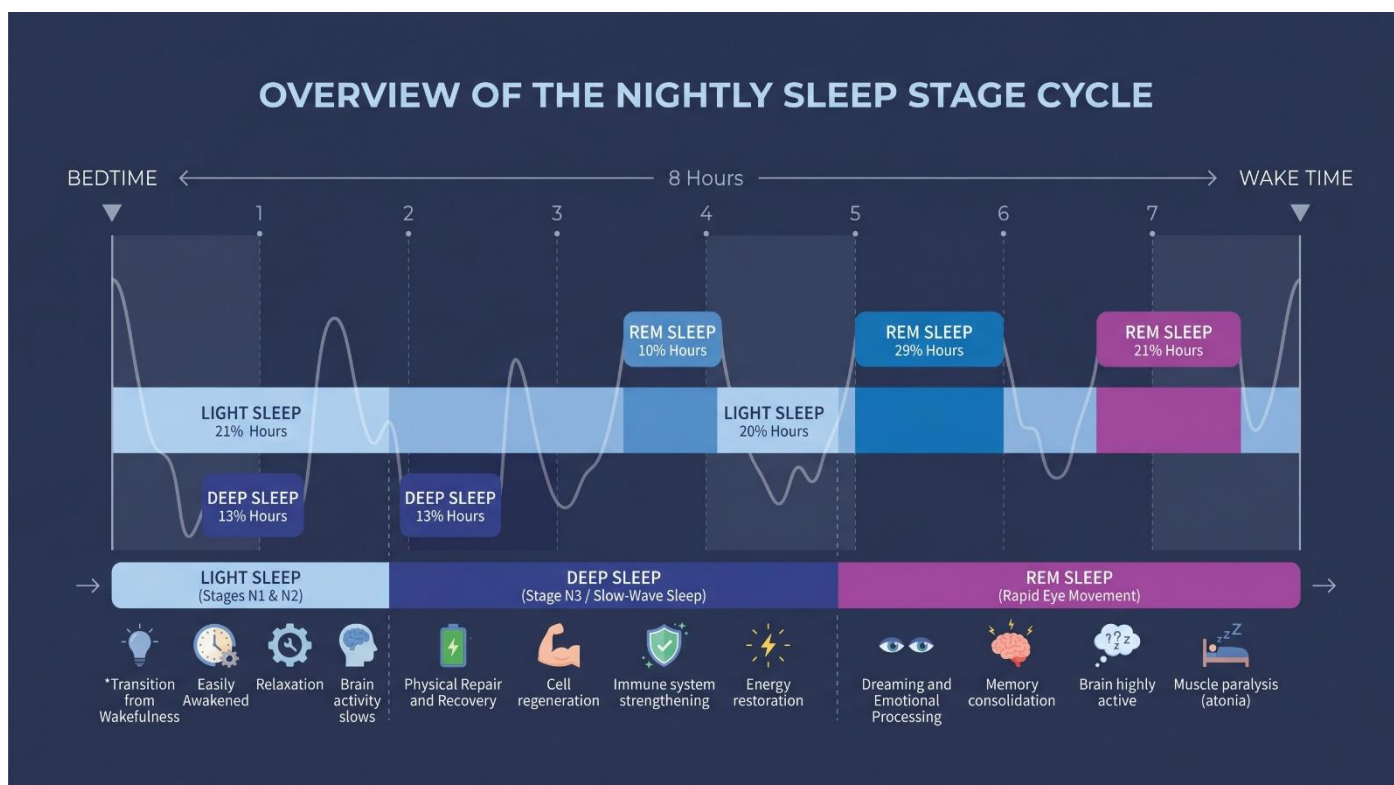
Nothing in this guide requires perfection or an expensive gadget. It requires curiosity: a willingness to experiment, notice what changes, and give your body a few weeks to respond. That's it.

THIS WEEK — TRY THIS

- ✓ For three consecutive days, note your wake-up time and rate how rested you feel on a 1–10 scale.
- ✓ Don't change anything yet. Just build a baseline.

2 What Deep Sleep Actually Does

Sleep is not passive. While you lie unconscious, your brain and body run through an intricate sequence of stages — each one doing work that waking life simply cannot do.



The Architecture of a Night

Scientists divide sleep into two broad types. Non-REM (NREM) sleep progresses from light (N1–N2) to deep slow-wave sleep (N3). REM sleep follows, characterised by rapid eye movements and brain activity that looks surprisingly similar to wakefulness. A typical night cycles through these stages roughly four to six times, each cycle lasting about 90 minutes.

Stage	Common Name	Key Functions
N1–N2	Light sleep	Transition into sleep; memory consolidation begins; easily disturbed.
N3	Deep / slow-wave	Physical repair; immune strengthening; growth hormone release; memory consolidation.
REM	Dream sleep	Emotional processing; creative problem-solving; procedural memory.

THE SCIENCE: Deep Sleep & Brain Cleansing

During N3 sleep, the brain's glymphatic system — a network of fluid-filled channels — becomes up to ten times more active than during waking hours. Cerebrospinal fluid flushes through the brain, clearing metabolic waste products including amyloid-beta and tau proteins implicated in Alzheimer's disease. This nightly 'wash cycle' is one of the strongest arguments for prioritising deep sleep quality, not just duration. (Xie et al., Science, 2013; Nedergaard & Goldman, Scientific American, 2016)

How Much Deep Sleep Do You Need?

Healthy adults typically spend 13–23% of their night in deep sleep — roughly 60–100 minutes across a 7–8 hour night. This proportion naturally declines with age: teenagers may spend over 20% in N3 while adults over 60 often average closer to 5–8%. Genetics also play a meaningful role.

The key insight: chasing a specific deep-sleep number is less useful than creating the conditions that allow your body to reach deep sleep naturally and consistently. This guide is about those conditions.

What Suffers When Deep Sleep Is Cut Short

Even one night of significantly reduced slow-wave sleep is detectable. Studies show impaired glucose regulation, elevated inflammatory markers, reduced growth hormone secretion, and weakened immune response. Chronically shallow sleep is associated with higher risks of cardiovascular disease, type 2 diabetes, and cognitive decline. Small, consistent improvements compound meaningfully over time.

THIS WEEK — TRY THIS

✓ If you use a sleep tracker, look only at total sleep time and wake time. Ignore stage percentages for now.

3 How Modern Life Quietly Undermines Sleep

Human sleep evolved in a world of firelight and dawn. We now live under LED screens, 24-hour artificial lighting, shift work, and the social expectation to be reachable at almost any hour. Each of these forces tugs at a different thread of your sleep architecture.

Light at the Wrong Times

Light is the primary signal that synchronises your circadian clock to the 24-hour day. It does this by suppressing melatonin — a hormone produced by the pineal gland that rises in darkness to initiate the biological cascade toward sleep. Blue-wavelength light (dominant in LED screens and overhead office lighting) suppresses melatonin most potently.

THE SCIENCE: Evening Light & Melatonin

A landmark Harvard study found that reading on a light-emitting device before bed suppressed melatonin by approximately 55% compared to reading a printed book, delayed the melatonin surge by 1.5 hours, delayed sleep onset, and reduced morning alertness even after 8 hours in bed. The study used devices at typical brightness — similar to what most people use nightly. (Chang et al., PNAS, 2015)

Irregular Schedules

Your circadian rhythm does not care that Saturday is a day off. When you sleep and wake at substantially different times on weekends versus weekdays — a phenomenon researchers call ‘social jet lag’ — you impose a repeated time-zone shift on your body. Deep sleep, which clusters preferentially in the first half of the night, can be systematically truncated. Research found social jet lag of just one hour was associated with measurably higher rates of obesity, mood disturbance, and metabolic dysfunction. (Roenneberg et al., Current Biology, 2012)

Caffeine

Caffeine works by blocking adenosine receptors. Adenosine is a chemical that accumulates in the brain throughout the day, building what is called sleep pressure. The longer you are awake, the more adenosine collects. Caffeine does not clear adenosine — it merely prevents you from feeling it. When the caffeine clears your system, the full adenosine load hits at once.

Caffeine’s half-life averages around five to six hours in most adults. A coffee at 3 p.m. still has half its caffeine circulating at 8–9 p.m. Research shows even moderate afternoon caffeine reduces slow-wave sleep significantly, even when subjects report no difficulty falling asleep. (Drake et al., Journal of Clinical Sleep Medicine, 2013)

Alcohol

Alcohol is the world’s most widely used sleep aid — and one of the most counterproductive. It does sedate the nervous system, making it easier to fall asleep. But as the liver metabolises alcohol, it produces a rebound effect that fragments sleep, suppresses REM sleep, and increases early-morning waking. Even moderate drinking (1–2 drinks) measurably reduces sleep quality by 9–24%. (Ebrahim et al., Alcoholism: Clinical & Experimental Research, 2013)

Chronic Stress

Prolonged stress keeps the HPA axis (hypothalamus–pituitary–adrenal stress system) in a state of low-level activation. Cortisol follows a natural daily curve — high in the morning, low at night. Chronic stress flattens that curve, sustaining elevated evening cortisol that directly suppresses slow-wave sleep. Racing thoughts at bedtime are both a symptom and a cause: they keep the nervous system in alert mode, preventing the shift toward parasympathetic dominance that deep sleep requires.

THIS WEEK — TRY THIS

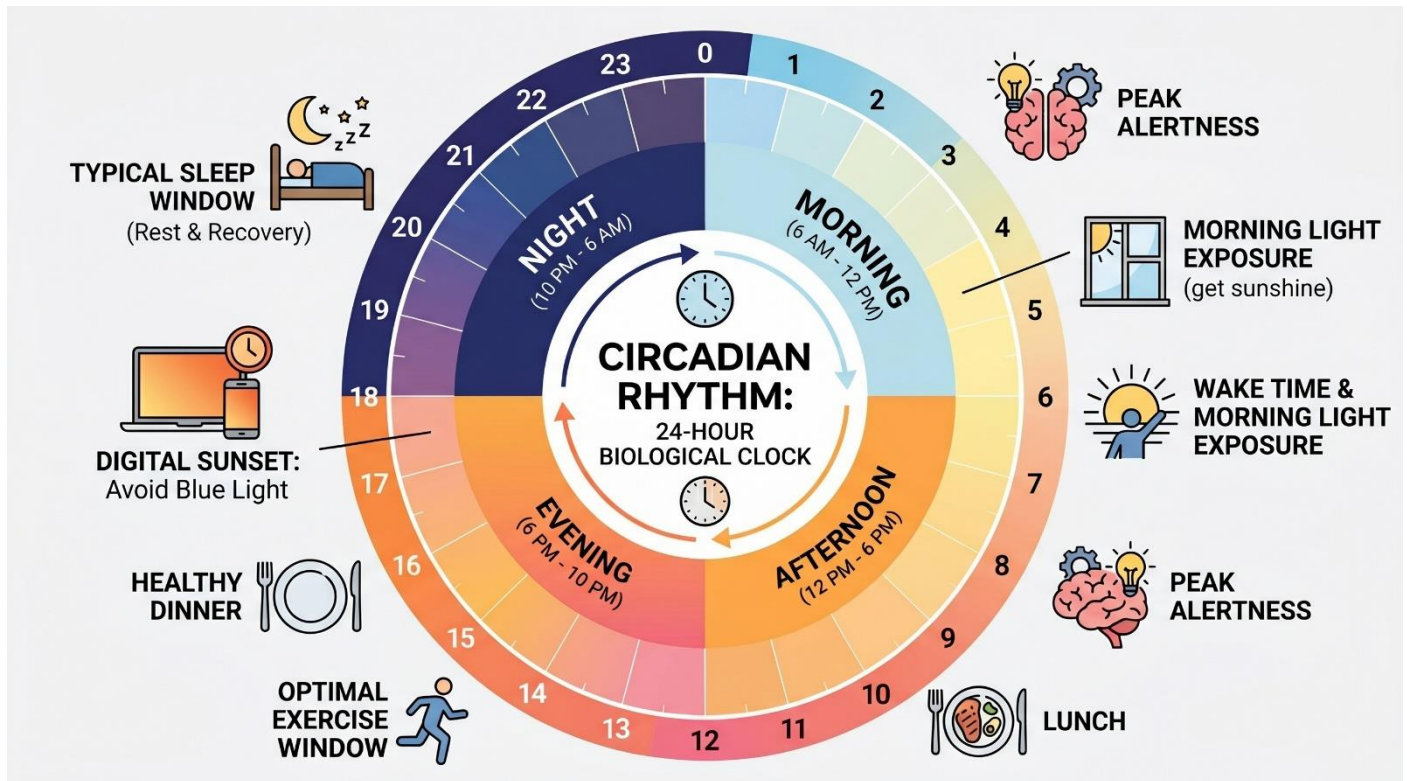
- ✓ Notice your screen and bright-light exposure in the two hours before bed.
- ✓ Note your last caffeine of the day and the time.
- ✓ Still just observing — changes come in the next chapter.

PART II

The Deep Sleep System

4 Set Your Clock: Light, Temperature & Timing

The four levers in this chapter are the foundation of everything else. Get these right and supplements, routines, and gadgets become refinements rather than crutches.



Morning Light: The Master Reset

Morning bright light sets your circadian rhythm and determines when melatonin will rise that evening. It signals the SCN (suprachiasmatic nucleus — your brain's master clock) to start a roughly 14–16 hour countdown to darkness. The earlier and brighter that signal, the more reliably your sleep drive builds and your melatonin window opens on time.

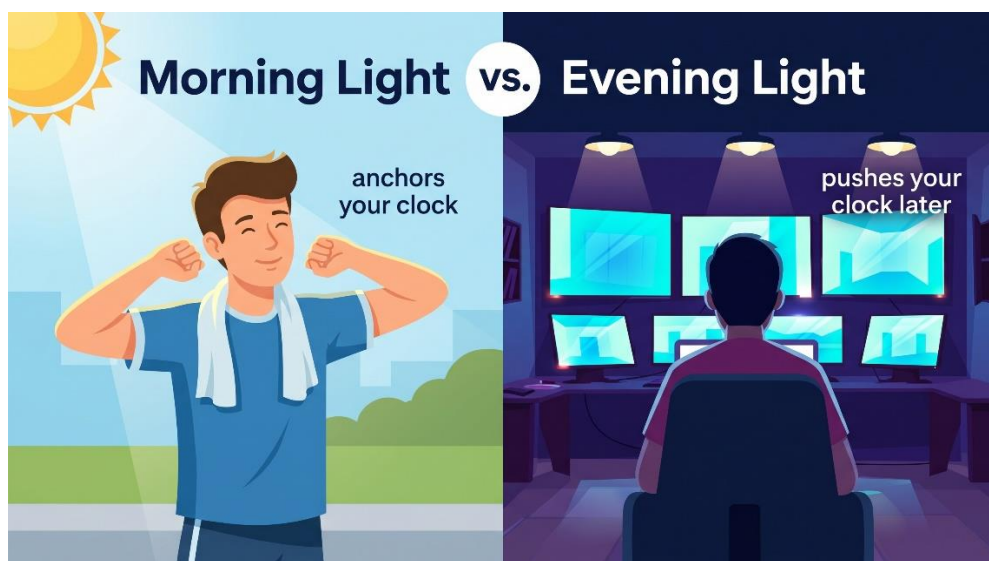
Outdoor light on a clear morning reaches 10,000–100,000 lux. Indoor office lighting typically delivers 200–500 lux — a factor of 20–200 times weaker. This is why even a brief outdoor exposure is orders of magnitude more effective than sitting by a window.

- ☐ Get outside within 30–60 minutes of waking.
- ☐ Aim for 5–10 minutes on bright days; 15–30 minutes when overcast.
- ☐ No sunglasses (ordinary prescription glasses and contacts are fine).
- ☐ If outdoor access is impossible, a 10,000-lux light therapy lamp for 20–30 minutes is an acceptable substitute.

Evening Light: The ‘Digital Sunset’

The evening goal is to mimic what your ancestors experienced after sunset: a gradual reduction in light intensity and a shift toward warmer light. This allows melatonin to rise naturally.

Time Before Bed	Action
2 hours	Dim overhead lights. Switch to lamps below eye level.
1 hour	Shift screens to night/warm mode, reduce brightness to ~50%.
30 min	Ideal: screens off. Read physical material, stretch, or talk.



Room Temperature

Sleep onset is partly triggered by a drop in core body temperature of about 1–2°C. Your bedroom should support this natural cooling.

- ☐ Target room temperature: 16–19°C (60–67°F). Experiment within this range.
- ☐ Use breathable natural-fibre bedding (cotton, linen, bamboo).
- ☐ A warm shower or bath 60–90 minutes before bed accelerates the cooling rebound — core temperature rises briefly, then drops sharply as you radiate heat from dilated skin vessels, speeding sleep onset.

The Single Most Powerful Schedule Change

If you do only one thing from this guide, do this: pick a fixed wake-up time and keep it within one hour, seven days a week. Not because sleep gurus say so, but because your circadian rhythm is a biological clock, and it runs best with a consistent reference point.

A stable wake time anchors the timing of your deepest sleep, regulates morning cortisol and melatonin timing, and builds adenosine pressure more predictably, making it easier to fall asleep the following night.

THIS WEEK — TRY THIS

- ✓ Choose a wake time you can hold even on weekends. Commit for 7 days.
- ✓ Get outside for morning light at least 5 of those 7 days.
- ✓ Start dimming your home 2 hours before bed.

5 Design Your Sleep Environment

Your bedroom should communicate a single message to your nervous system: this is where we rest. Not where we work, scroll, or process the day's stress. The most effective changes are often the cheapest.

Darkness

Even dim light during sleep suppresses melatonin and activates alerting brain circuits. Research shows sleeping in even moderate room light increases heart rate and insulin resistance the following morning. (Mason et al., PNAS, 2022)

- ☐ Blackout curtains or a well-fitted eye mask are the goal.
- ☐ Cover or tape over LED indicator lights on chargers, routers, and TVs.
- ☐ If you use a clock, face it away from you.

Sound

Noise disrupts sleep even when it does not fully wake you — the brain continues processing sound during N2 and REM sleep, and intermittent unexpected sounds produce micro-arousals that fragment architecture without reaching consciousness.

- ☐ Rugs absorb sound; heavy curtains dampen street noise; foam earplugs block most environmental sound.
- ☐ White noise or pink noise masks irregular sound spikes. A fan, a dedicated white-noise machine, or a free app all work equally well.

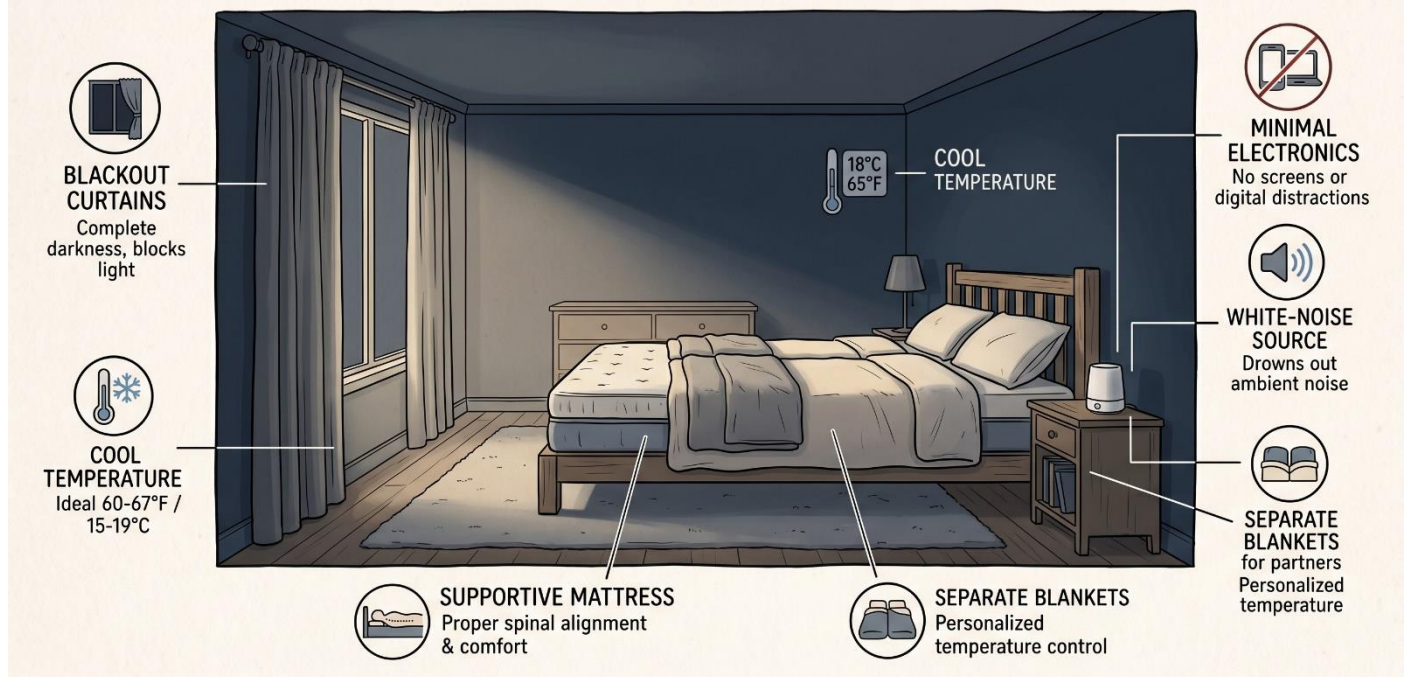
The Bed Rule

Stimulus control is one of the most evidence-based behavioural interventions for sleep. If you regularly do alerting or stressful activities in bed — working, scrolling news, watching emotionally intense content — your brain learns to associate the bed with wakefulness. Use your bed only for sleep and intimacy. It sounds almost too simple. It works.

SLEEP ENVIRONMENT CHECKLIST

- ☐ Room temperature feels cool (16–19°C / 60–67°F)
- ☐ Blackout curtains or eye mask in place
- ☐ LED indicator lights covered
- ☐ Noise low or masked by white/pink noise
- ☐ Mattress and pillow support comfortable spinal alignment
- ☐ No screens used in bed for work or social media
- ☐ Phone on silent / do-not-disturb (or in another room)

“SLEEP CAVE” BEDROOM CHECKLIST



THIS WEEK — TRY THIS

- ✓ Make one environment change tonight (darkness is the highest-leverage start).
- ✓ Plan one more change for later this week.

6 Wind-Down Rituals That Actually Work

You can't sprint to the end of the runway and expect to glide smoothly to a stop. The nervous system needs a genuine transition from 'engaged' to 'at rest' — and that transition takes time.

What's Actually Happening During Wind-Down

During the wind-down window, you are trying to shift autonomic balance from sympathetic dominance (elevated heart rate, cortisol, alerting noradrenaline) toward parasympathetic dominance (lower heart rate, falling core temperature, rising melatonin). Activities that are cognitively engaging, emotionally stimulating, or involve bright screens delay this shift — even if the content seems nominally relaxing.

Tools That Work

Physical reading in low light

Fiction or non-urgent non-fiction. Avoids the reactive loop of news and social media. The physical book aspect matters — no screen light, no notification pull.

Gentle movement or stretching

Light yoga or stretching activates the parasympathetic nervous system through slow, controlled breathing and reduces residual muscle tension from the day.

Non-Sleep Deep Rest (NSDR) / body scan

Guided relaxation practices that direct attention to physical sensations rather than thoughts. Research from Stanford shows NSDR can restore neurochemical reserves and reduce sleep-onset anxiety even on nights where full sleep is difficult.

Brain dump / worry list

A 5–10 minute writing exercise where you externalise tomorrow's tasks and unresolved concerns onto paper. Designated 'worry time' earlier in the evening significantly reduces intrusive bedtime thoughts.

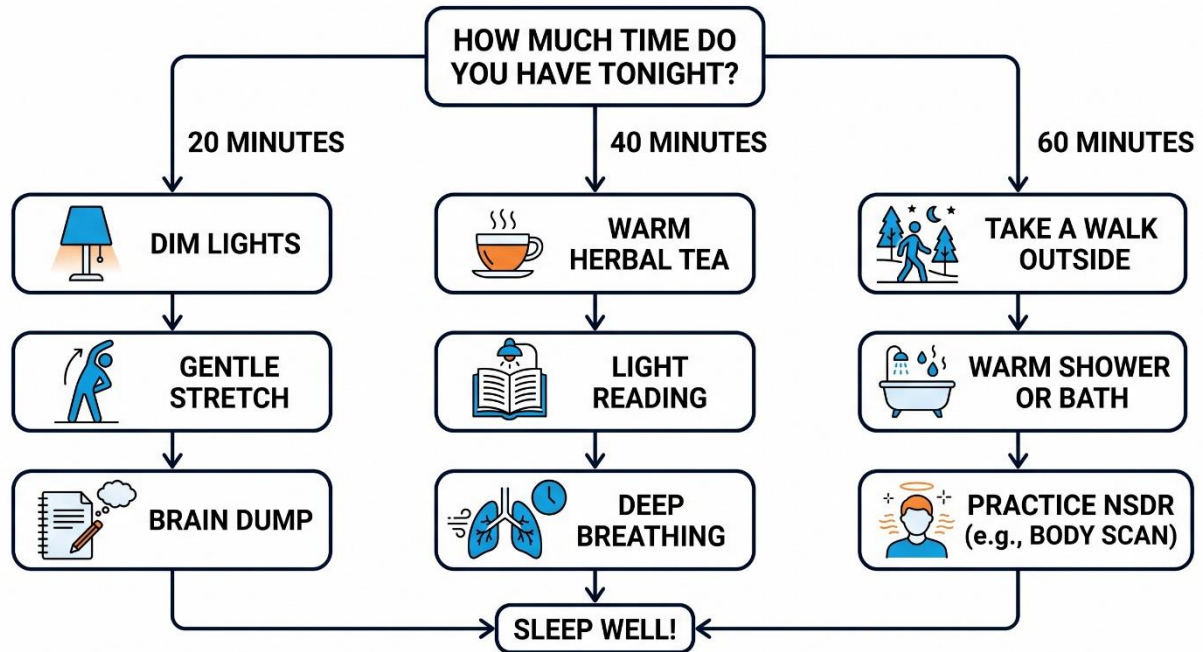
Three Practical Templates

Template	Steps
20-Min 'Busy Night'	Dim overhead lights (2 min) → Light stretching or 4-7-8 breathing (10 min) → Write tomorrow's top 3 priorities (5 min) → Brief body scan (3 min)
40-Min 'Standard'	Close work apps; tidy small things (10 min) → Read something light (15 min) → Guided NSDR or body scan audio (15 min)
60-Min 'Full Reset'	Gentle walk or movement (15 min) → Warm shower (15 min) → Journalling or gratitude writing (15 min) → NSDR or relaxation audio (15 min)

THIS WEEK — TRY THIS

- ✓ Choose one template and try it three nights this week.
- ✓ Adjust timing to fit your life, but keep the basic shape.

CHOOSE YOUR WIND-DOWN ROUTINE



7 Food, Protein & Supplements

Nutrition's impact on sleep is real but secondary — it works best once the foundations of light, timing, and environment are in place. Think of this chapter as fine-tuning, not a shortcut.

Meal Timing

Digestion is metabolically demanding. A large meal within 2–3 hours of bed elevates core body temperature (the opposite of what sleep requires), increases the likelihood of acid reflux, and can trigger blood-glucose fluctuations that fragment sleep in the second half of the night. Finishing your main meal at least 3 hours before bed is a consistent recommendation across sleep medicine guidelines.

WHICH PROTEIN BEFORE BED?

CASEIN PROTEIN	WHEY ISOLATE	WHEY CONCENTRATE/BLEND
 • SLOW RELEASE (e.g., 6–8 hrs)	 • FAST DIGESTING (e.g., 30–60 min)	 • MODERATE DIGESTION (e.g., 1–2 hrs)
 **OVERNIGHT MUSCLE RECOVERY** • PREVENT BREAKDOWN	 • **POST-WORKOUT REPAIR** • QUICK AMINO BOOST	 **GENERAL DAY USE**, MEAL REPLACEMENT • BUDGET-FRIENDLY
Typical Serving Range  20–40 G (Per Serving)	Typical Serving Range  20–40 G (Per Serving)	Typical Serving Range  20–40 G (Per Serving)

CLEAN FOOTER

Pre-Sleep Protein

For people who strength train or want better overnight recovery, pre-sleep protein supplementation has a meaningful evidence base. A study found 40 g of casein protein before bed increased overnight muscle protein synthesis by approximately 22% in trained men. Subsequent research has replicated the benefit with doses as low as 20–30 g. (Res et al., Medicine & Science in Sports & Exercise, 2012)

Type	Key Properties	Best For
Casein	Slow-digesting (7–8 hr); forms a gel in the stomach	Muscle recovery; longer overnight amino acid supply
Whey isolate	Fast-digesting (1–2 hr); high leucine	Lactose sensitivity; lighter pre-bed option
Plant blend (pea + rice)	Medium absorption; complete amino acid profile	Dairy-free; vegan

Key Supplements

Supplement	Mechanism	Dose	Timing	Notes
Magnesium glycinate/threonate	Activates GABA receptors; reduces cortisol; supports slow-wave sleep	200–400 mg elemental Mg	1–2 hrs before bed	Oxide form poorly absorbed; glycinate well-tolerated; threonate may add cognitive benefits
L-Theanine	Promotes alpha brain waves; reduces anxiety without sedation	100–200 mg	30–60 min before bed	Safe to combine with magnesium; may reduce sleep-onset time
Melatonin	Signals darkness onset to the SCN; does not induce sleep directly	0.1–0.5 mg (low dose)	30–60 min before target bedtime	Most OTC doses (3–10 mg) far exceed physiological levels. Low dose more effective for circadian shifting
Glycine	Lowers core body temperature; may reduce daytime fatigue after poor sleep	3 g	Before bed	Generally well tolerated; early evidence suggests improved subjective sleep quality

THIS WEEK — TRY THIS

- ✓ Move your main meal to at least 3 hours before bed if it's currently later.
- ✓ Set a caffeine cut-off (a good starting point: no caffeine after 2 p.m.).
- ✓ If considering supplements, start with one — magnesium glycinate is a reasonable first choice — and trial it for at least 7 nights before judging.

PART III

Mind, Data & Long-Term Change

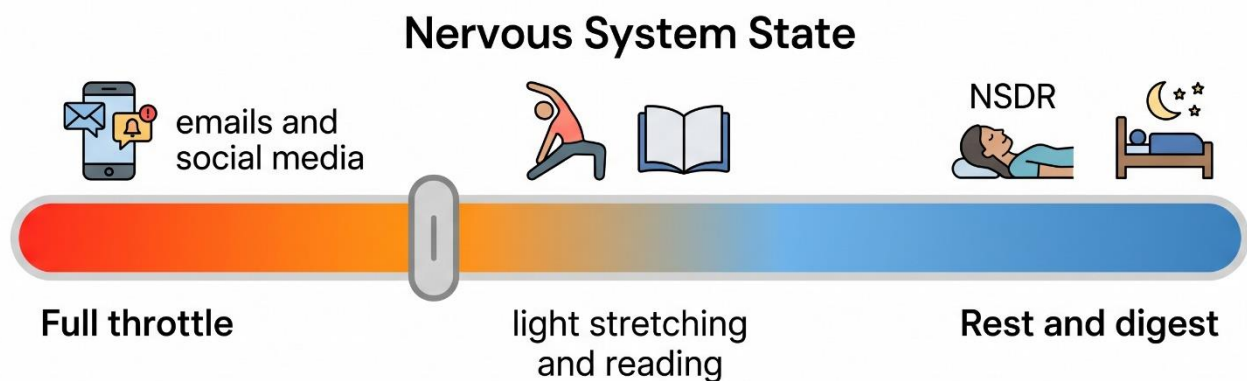
8 Calm the ‘Wired but Tired’ Mind

Perhaps the most frustrating sleep experience: you’re exhausted, you’ve done everything right, you lie down — and your brain decides this is an excellent time to review every unresolved problem from the past six months.

Why Trying Harder Makes It Worse

Sleep is a paradoxical process — the harder you try to force it, the more elusive it becomes. The clinical term is ‘sleep effort’: the degree of cognitive and emotional work directed toward making yourself sleep. Sleep effort activates the arousal system, raising heart rate and cortisol, directly opposing the physiological conditions needed for sleep onset. This is why people with chronic insomnia often show hyperarousal — measurably higher core body temperature, heart rate, and metabolic rate at night — despite feeling exhausted. (Perlis et al., *Sleep Medicine Reviews*, 1997)

Why We Need a Wind-Down.



Evidence-Based Mind Tools

Stimulus Control Therapy

One of the most replicated non-drug interventions for insomnia. Use the bed only for sleep and sex; go to bed only when sleepy (not just tired); if you cannot sleep after approximately 20 minutes, get up and do something calm in dim light until sleepy again; maintain a consistent wake time regardless of how the night went. By repeatedly pairing the bed with sleepiness rather than wakefulness, you rebuild the automatic association over one to two weeks.

Scheduled Worry Time

Set aside 15–20 minutes earlier in the evening — not in bed, not in the last hour before sleep — to write down current worries and the next small action for each one. When a thought appears at bedtime, you can genuinely tell yourself: ‘I have a time for this. It’s handled enough for tonight.’

Cognitive Defusion

From Acceptance and Commitment Therapy (ACT): rather than fighting thoughts or trying to replace them, observe them as mental events without engaging their content. ‘There’s the work worry again’ rather than ‘I need to solve this right now.’ This reduces emotional reactivity without requiring thought suppression, which reliably backfires.

NSDR and the Physiological Sigh

Non-Sleep Deep Rest practices (yoga nidra, body scans, guided relaxations) consistently show measurable reductions in cortisol and subjective anxiety even in a single session. For rapid nervous system downregulation, the ‘physiological sigh’ — a double inhale through the nose followed by a long exhale through the mouth — is the fastest single-breath technique to activate the parasympathetic branch. (Balban et al., Cell Reports Medicine, 2023)

THIS WEEK — TRY THIS

- ✓ Choose one tool from this chapter and practise it 3–5 nights — even on nights when sleep feels fine, so your brain learns the pattern.
- ✓ If you find yourself clock-watching, turn the clock face away.

9 Using Trackers Without Obsessing

Wearables and sleep apps can be genuinely useful. They can also become a source of anxiety that makes your sleep worse. A 2017 paper in the *Journal of Clinical Sleep Medicine* coined the term ‘orthosomnia’ for the phenomenon of people developing sleep anxiety specifically from pursuing better tracker scores.

What Trackers Do Well

- Revealing trends in total sleep time over weeks.
- Showing how consistent your bedtime and wake time actually are versus how consistent you think they are.
- Detecting correlations between habits and outcomes — e.g., that late alcohol reliably degrades your resting heart rate the following night.

What Trackers Do Poorly

Consumer wearables infer sleep stages from movement and heart rate variability — not from the brain activity (EEG) used in clinical sleep labs. Validation studies consistently show that devices are reasonable at detecting total sleep time (within 10–15%) but substantially less accurate at staging. A device might report 45 minutes of deep sleep when a simultaneous clinical study records 80 minutes. Use stage data as a rough directional signal over weeks, not a nightly diagnosis.

A Saner Approach to Data

Metric	How to Use It
Total sleep time	Track nightly. Weekly average is most meaningful.
Bedtime consistency	Aim for ≤ 30 -min variation. Wider = investigate why.
Wake time consistency	Your anchor. Protect this above all else.
RHR / HRV (if available)	Useful proxy for recovery. Trends matter, not single nights.
Deep / REM percentages	Interesting directional data only. Do not optimise directly.

THIS WEEK — TRY THIS

- ✓ Designate one day per week (e.g., Sunday morning) as your data review day.
- ✓ On all other days, check total sleep time only — then close the app.

10 Troubleshooting, Plateaus & When to Get Help

Progress with sleep is rarely linear. You will have rough stretches — travel, illness, work pressure, a newborn, grief. The goal is not to never have a bad night; it's to have a system you can return to.

Quick Troubleshooting Path

Still taking more than 30 minutes to fall asleep

Check caffeine cut-off, evening light, and whether you're going to bed before you're genuinely sleepy. Consider getting up at the 20-minute mark rather than lying awake (stimulus control).

Waking at 2–3 a.m. and unable to return to sleep

Common causes: alcohol (even 1–2 drinks), late eating, or high evening cortisol. Also check room temperature — waking in the early hours is often associated with the body warming as morning approaches.

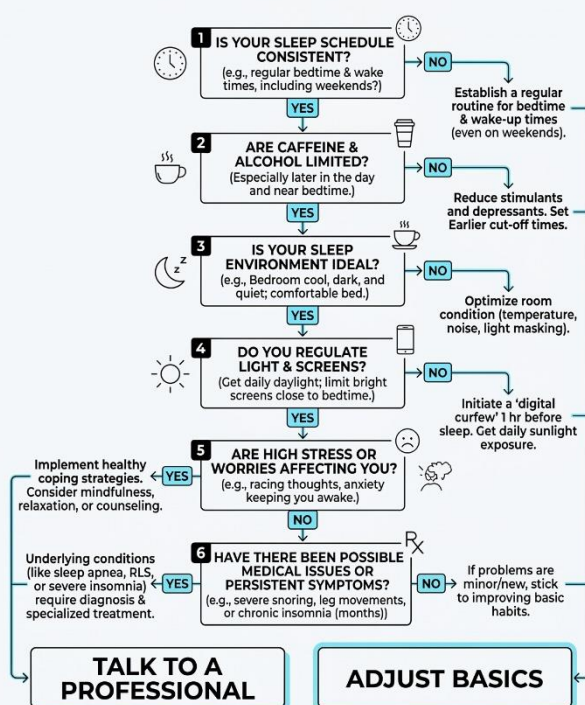
Waking exhausted despite 8+ hours in bed

This pattern — adequate duration, poor restorativeness — is a classic flag for sleep apnea, which affects an estimated 1 billion people worldwide and is severely underdiagnosed. Seek medical evaluation.

Hitting a plateau after initial improvement

Normal. Revisit the Week 1 foundations (light and timing) first — they slip gradually. Only after basics are solid should you add new interventions.

STILL NOT SLEEPING WELL? START HERE.



When to Seek Professional Help

This guide is for people with lifestyle-based sleep difficulties. Some conditions require professional diagnosis and treatment that no habit change will replace. Speak to a health professional if:

- ☐ You snore loudly, wake gasping, or have been told you stop breathing during sleep.
- ☐ You feel dangerously sleepy while driving or in safety-critical situations.
- ☐ Restless, uncomfortable sensations in your legs disrupt sleep (possible Restless Legs Syndrome).
- ☐ You have insomnia at least 3 nights per week for more than 3 months, despite consistent implementation of these steps.
- ☐ You feel persistently low, hopeless, or anxious alongside poor sleep — these may be cause rather than consequence, and are treatable.

Cognitive Behavioral Therapy for Insomnia (CBT-I) is the gold-standard first-line treatment for chronic insomnia according to the American Academy of Sleep Medicine — more effective than sleeping medication in the long term, with no side effects. It is available with a therapist, in group format, or through several validated digital programs.

THIS WEEK — TRY THIS

- ✓ Note any recurring issues not resolving after 3–4 weeks of consistent effort.
- ✓ If anything from the list above applies to you, book that appointment.

PART TOOLS

Appendices

A 28-Day Deep Sleep Action Plan

Week	Focus	Daily Actions
Week 1 Light & Timing	Anchor your clock	Day 1: Choose and commit to a fixed wake time. Days 2–5: Morning outdoor light within 60 min of waking every day. Days 6–7: Begin dimming home 2 hours before bed.
Week 2 Environment & Routine	Optimise your sleep space	Day 8: Add blackout solution to bedroom. Day 9: Remove/cover LED indicators; phone out of bed. Days 10–14: Trial one wind-down routine (20, 40, or 60 min) at least 4 nights.
Week 3 Food & Caffeine	Adjust nutrition timing	Day 15: Move main meal to at least 3 hours before bed. Day 16: Set a caffeine cut-off (start with 2 p.m.). Days 17–21: Observe energy, mood, and sleep quality. Adjust if needed.
Week 4 Supplements & Data	Fine-tune and review	Day 22: If appropriate, choose one supplement (magnesium glycinate recommended). Days 23–26: Trial chosen supplement for at least 7 nights. Days 27–28: Full weekly data review. Identify 1–2 changes with clearest positive effect and build those in permanently.

B Key References & Further Reading

The following sources underpin the claims made throughout this guide.

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Orthosomnia

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Qaseem A et al. Management of chronic insomnia disorder in adults. Annals of Internal Medicine 2016;165:125–133.

For accessible further reading: *Why We Sleep* by Matthew Walker (2017) provides a broad overview of sleep science for general audiences.

The *Sleep Solution* by W. Chris Winter (2017) is a more clinical and practical companion. For CBT-I specifically, *Overcoming Insomnia* by Jack Edinger and Colleen Carney is the therapist-guide standard.